

NUMERICAL ANALYSIS

PROBLEM SET 4

ROB TOMPKINS

1. PROBLEM 1

1.a. We are deriving the 2-point open Newton-Coats formula

$$\int_a^b f(x)dx \approx I_{2,\text{open}}(f) = \frac{b-a}{3} \left(2f(a+h) - f(a+2h) + 2f(a+3h) \right)$$

for evenly spaced points with $h = \frac{b-a}{n+2}$. Now we set $n = 2$ so that $h = \frac{b-a}{4}$, and we need to sort out

$$(1) \quad I_{2,\text{open}}(f) = \int_a^b \sum_{i=0}^2 L_{2,i}(x) f(x_i) dx.$$

But we need to shift to an easier or simpler variable to work with. So we set $x = a + t \cdot h$, with $dx = h \cdot dt$. Now as x moves from a to b we see that ($b = a + 4h$), t moves from 0 to 4, and we have

$$\int_a^b f(x)dx = h \int_0^4 f(a+ht)dt,$$

which we interpolate $f(a+ht)$ at $t = 1, 2, 3$. We write out the Lagrangian polynomials $L_{2,0}, L_{2,1}, L_{2,2}$ for the sake of the problem. This follows as

$$L_{2,0}(x) = \frac{(t-2)(t-3)}{(1-2)(1-3)} = \frac{(t-2)(t-3)}{2} = \frac{1}{2}(t^2 - 5t + 6)$$

$$L_{2,1}(x) = \frac{(t-1)(t-3)}{(2-1)(2-3)} = -(t-1)(t-3) = -t^2 + 4t - 3$$

$$L_{2,2}(x) = \frac{(t-1)(t-2)}{(3-1)(3-2)} = \frac{(t-1)(t-2)}{2} = \frac{1}{2}(t^2 - 3t + 2).$$

And we then need to integrate these Lagrange interpolants, which we do as

$$\begin{aligned}\int_0^4 \frac{1}{2}(t^2 - 5t + 6)dt &= \frac{t^3}{6} - \frac{5t^2}{4} + 3t \Big|_0^4 = \frac{8}{3} \\ \int_0^4 -t^2 + 4t - 3dt &= -\frac{t^3}{3} + 2t^2 - 3t \Big|_0^4 = -\frac{4}{3} \\ \int_0^4 \frac{1}{2}(t^2 - 3t + 2)dt &= \frac{t^3}{6} - \frac{3t^2}{2} + 2t \Big|_0^4 = \frac{8}{3}\end{aligned}$$

Thus, when we plug this into equation (1), we get the following

$$\begin{aligned}\int_a^b f(x)dx &\approx h \left(\frac{8}{3}f(a+h) - \frac{4}{3}f(a+2h) + \frac{8}{3}f(a+3h) \right) \\ &\approx \frac{4h}{3} \left(2f(a+h) - f(a+2h) + 2f(a+3h) \right) \\ &\approx \frac{b-a}{3} \left(2f(a+h) - f(a+2h) + 2f(a+3h) \right).\end{aligned}$$

which solves our problem

1.b. We must show that our quadrature is exact for $1, x, x^2, x^3$ but not for x^4 . Again we rescale the interval in teh fashion

$$x = a + ht, \quad h = \frac{b-a}{4}, x \in [a, b] \rightarrow t \in [0, 4]$$

So $Q(g) = \frac{4}{3} \left(2g(1) - g(2) - 2g(3) \right)$. But, we need to check if $I(g) = Q(g)$ for $g(t) \in \{1, x, x^2, x^3, x^4\}$. So we write

$$\begin{aligned}\int_0^4 1dt &= 4 \stackrel{?}{=} \frac{4}{3}(2 - 1 + 2) = 4 \quad \checkmark \\ \int_0^4 tdt &= \frac{t^2}{2} \Big|_0^4 = 8 \stackrel{?}{=} \frac{4}{3}(2 \cdot 1 - 2 + 2 \cdot 3) = 8 \quad \checkmark \\ \int_0^4 t^2dt &= \frac{t^3}{3} \Big|_0^4 = \frac{64}{3} \stackrel{?}{=} \frac{4}{3}(2 \cdot 1 - 4 + 2 \cdot 9) = \frac{4 \cdot 16}{3} = \frac{64}{3} \quad \checkmark \\ \int_0^4 t^3dt &= \frac{t^4}{4} \Big|_0^4 = 64 \stackrel{?}{=} \frac{4}{3}(2 \cdot 1 - 8 + 2 \cdot 27) = \frac{4 \cdot 48}{3} = \frac{192}{3} = 64 \quad \checkmark \\ \int_0^4 t^4dt &= \frac{t^5}{5} \Big|_0^4 = \frac{1024}{5} = 204.8 \stackrel{?}{=} \frac{4}{3}(2 - 32 + 2 \cdot 243) = \frac{1824}{3} = 608 \quad \times\end{aligned}$$

So we are done.

2. PROBLEM 2

Let

$$I(f) = \int_a^b f(x)dx$$

and let $[a, b]$ be divided into subintervals in the following fashion

$$\{[a, x_1 + h], [x_2 - h, x_2 + h], \dots, [x_n - h, b]\},$$

with $x_1, \dots, x_n$ being the midpoints of the subintervals. We thusly see that $h = \frac{b-a}{2n}$, that $x_j = a + (2j-1)h$, and that each of our subintervals has width $2h$. Now consider the j^{th} subinterval $[x_j - h, x_j + h]$. We can write $f(x)$ over the interval using Taylor's theorem as

$$f(x) = f(x_j) + f'(x_j)(x - x_j) + \frac{f''(\eta_x)}{2}(x - x_j)^2,$$

for some η_x between x and x_j . So we integrate

$$\begin{aligned} \int_{x_j-h}^{x_j+h} f(x)dx &= \int_{x_j-h}^{x_j+h} \left(f(x_j) + f'(x_j)(x - x_j) + \frac{f''(\eta_x)}{2}(x - x_j)^2 \right) dx \\ &= \int_{x_j-h}^{x_j+h} f(x_j)dx + \int_{x_j-h}^{x_j+h} f'(x_j)(x - x_j)dx + \int_{x_j-h}^{x_j+h} \frac{f''(\eta_x)}{2}(x - x_j)^2 dx \\ &= 2hf(x_j) + 0 + \frac{f''(\xi_j)}{2} \int_{x_j-h}^{x_j+h} (x - x_j)^2 dx, \end{aligned}$$

for some ξ_j clearly dependent upon our choice of η_x . But notice that our last integral can be manipulated by a transformation of variables in the following fashion

$$\int_{x_j-h}^{x_j+h} (x - x_j)^2 dx = \int_{-h}^h t^2 dt = \frac{2h^3}{3}.$$

Substituting that back in to the over all equation we see that

$$\int_{x_j-h}^{x_j+h} f(x)dx + \frac{f''(\xi_j)}{2} \frac{2h^3}{3}.$$

So we take this result and sum across all the subintervals to get that

$$\int_b^a f(x)dx = \sum_{j=1}^n \int_{x_j-h}^{x_j+h} f(x)dx = 2h \sum_{j=1}^n f(x_j) + \frac{h^3}{3} \sum_{j=1}^n f''(\xi_j).$$

But we notice that $\sum_{j=1}^n f''(\xi_j) = n f''(\xi)$ with $f''(\xi)$ representing the average of all $f''(\xi_j)$. And with some careful substitution, along with noticing that $n = \frac{b-a}{2h}$ by manipulating our definition of h , we see that

$$\int_a^b f(x)dx = 2h \sum_{j=1}^n f(x_j) + \frac{(b-a)h^2}{6} f''(\xi),$$

where $\xi \in [a, b]$, and we are done.

3. PROBLEM 3

We will determine the weights w_i and the quadrature points x_i in the three point Gauss-Legendre quadrature formula

$$\int_{-1}^1 f(x)dx \approx w_1 f(x_1) + w_2 f(x_2) + w_3 f(x_3).$$

Now by definition the degree of precision of this formula just falls out by us having to generate $2n - 1$ equations, so that we can use the method of undetermined coefficients for finding the weights. This yields a $2(3) - 1 = 5$ degree of precision. So we go through the process of generating the equations over which we'll solve for the weights by using the integrals of $1, x, x^2, x^3, x^4, x^5$, as

$$\begin{aligned} f(x) = 1 : \quad w_1 + w_2 + w_3 &= \int_{-1}^1 1 \cdot dx = 2, \\ f(x) = x : \quad w_1 x + w_2 x + w_3 x &= \int_{-1}^1 x dx = \frac{x^2}{2} \Big|_{-1}^1 = 0, \\ f(x) = x^2 : \quad w_1 x^2 + w_2 x^2 + w_3 x^2 &= \int_{-1}^1 x^2 dx = \frac{x^3}{3} \Big|_{-1}^1 = \frac{2}{3}, \\ f(x) = x^3 : \quad w_1 x^3 + w_2 x^3 + w_3 x^3 &= \int_{-1}^1 x^3 dx = \frac{x^4}{4} \Big|_{-1}^1 = 0, \\ f(x) = x^4 : \quad w_1 x^4 + w_2 x^4 + w_3 x^4 &= \int_{-1}^1 x^4 dx = \frac{x^5}{5} \Big|_{-1}^1 = \frac{2}{5}, \\ f(x) = x^5 : \quad w_1 x^5 + w_2 x^5 + w_3 x^5 &= \int_{-1}^1 x^5 dx = \frac{x^6}{6} \Big|_{-1}^1 = 0. \end{aligned}$$

Our goal now is to define a polynomial, call it $p(x)$, that vanishes at the quadrature points x_1, x_2, x_3 . So we have

$$\begin{aligned} p(x) &= (x - x_1)(x - x_2)(x - x_3) \\ &= x^3 - x_1 x^2 - x_2 x^2 - x_3 x^2 + x_1 x_2 x + x_1 x_3 x + x_2 x_3 x - x_1 x_2 x_3 \\ &= x^3 s_1 x^2 + s_2 x + s_3, \end{aligned}$$

where $s_1 = -(x_1 + x_2 + x_3)$, $s_2 = x_1 x_2 + x_1 x_3 + x_2 x_3$, and $s_3 = -x_1 x_2 x_3$.

Now to find our weights we set $p(x) = 0$, then $xp(x) = 0$, and finally $x^2p(x) = 0$. So let us begin with

$$\begin{aligned} p(x_i) = 0 &= x_i^3 + s_1x_i^2 + s_2x_i + s_3 = 0 \\ &= \sum w_ix_i^3 + s_1 \sum w_ix_i^2 + s_2 \sum w_ix_i + s_3 \sum w_i = 0 \\ &= 0 + s_1 \frac{2}{3} + 0 + s_3 \cdot (-2) \end{aligned}$$

So we have that

$$s_1 = -3s_3.$$

Now set $xp(x) = 0$, and let us see what we can find:

$$\begin{aligned} xp(x) &= x^4 + s_1x^3 + s_2x^2 + s_3x = 0 \\ &= \sum w_ix_i^4 + s_1 \sum w_ix_i^3 + s_2 \sum w_ix_i^2 + s_3 \sum w_ix_i = 0 \\ &= \frac{2}{5} + s_1 \cdot 0 + s_2 \frac{2}{3} + s_3 \cdot 0 = 0 \\ &= \frac{2}{5} + s_2 \frac{2}{3} = 0 \quad \Rightarrow \quad s_2 = -\frac{3}{5}. \end{aligned}$$

Next we set $x^2p(x) = 0$ to again have another equation over which we can solve for s_1, s_2, s_3 , which follows as

$$\begin{aligned} x^2p(x) &= x^5 + s_1x^4 + s_2x^3 + s_3x^2 = 0 \\ &= \sum w_ix_i^5 + s_1 \sum w_ix_i^4 + s_2 \sum w_ix_i^3 + s_3 \sum w_ix_i^2 = 0 \\ &= 0 + s_1 \frac{2}{5} + s_2 \cdot 0 + s_3 \frac{2}{3} = 0 \\ &= s_1 \frac{2}{5} + s_3 \frac{2}{3}. \end{aligned}$$

Because we have $s_1 = -3s_3$ we see that $-6s_3/5 + 2s_3/3 = s_3(2/3 - 6/5) = 0$. So, $s_3 = 0$ and thus $s_1 = 0$, which setting $p(x) = 0$ yields

$$p(x) = x^3 - \frac{3}{5}x = x \left(x^2 - \frac{3}{5} \right) = 0.$$

So we see that

$$x_1 = -\sqrt{\frac{3}{5}}, \quad x_2 = 0, \quad x_3 = \sqrt{\frac{3}{5}}.$$

So now we solve for the weights using our first three $f(x)$ equations from the beginning. We begin with,

$$(2) \quad \sum w_i = w_1 + w_2 + w_3 = 2.$$

Next, we see that

$$\sum w_i x_i = -w_1 \sqrt{\frac{3}{5}} + w_2 \cdot 0 + w_3 \sqrt{\frac{3}{5}} = 0 \quad \Longleftrightarrow \quad w_1 = w_3.$$

Lastly we compute,

$$\sum w_i x_i^2 = w_1 \frac{3}{5} + w_3 \frac{3}{5} = \frac{6w_1}{5} = \frac{2}{3} \quad \Longleftrightarrow \quad w_1 = w_3 = \frac{5}{9}$$

Which plugging back into equation (2) yields

$$\frac{5}{9} + w_2 + \frac{5}{9} = 2 \quad \Longleftrightarrow \quad w_2 = \frac{8}{9}.$$

So we see that

$$\int_{-1}^1 f(x) dx \approx \frac{5}{9} f\left(-\sqrt{\frac{3}{5}}\right) + \frac{8}{9} f(0) + \frac{5}{9} f\left(\sqrt{\frac{3}{5}}\right),$$

and we are done.

4. PROBLEM 4

Consider the initial value problem given as

$$y' = y \cos(t), \quad y(0) = 1, \quad t \in [0, 2\pi].$$

4.a. We need to show existence and uniqueness of the solution of some subinterval $I \subseteq [0, 2\pi]$ containing $t = 0$. To do this we show the Lipschitz condition $|f(t, y_1) - f(t, y_2)| \leq L|y_1 - y_2|$. First we know that $f(x, y)$ must be continuous in our domain because it is the product of two continuous functions. Next we notice that

$$|f(t, y_1) - f(t, y_2)| = |y_1 \cos(t) - y_2 \cos(t)| = |\cos(t)| |y_1 - y_2| \leq 1 \cdot |y_1 - y_2|,$$

because $|\cos(t)| \leq 1$ for all $t \in \mathbb{R}$. So our function is Lipschitz with $L = 1$. Furthermore, we have a theorem that states that if we are Lipschitz on our domain then for an initial value problem of our form, then we are guaranteed existence and uniqueness of the solution over our interval.

4.b. We now compute the Forward Euler Method for solving the problem. Specifically, we need to work out the stepwise solution for $y_{n+1} = y_n + hf(t, y_n)$. Set $h = \pi/4$ with $t_0 = 0$, $y_0 = 1$, $t_1 = \pi/4$, and $t_2 = \pi/2$.

Step 1. With $t_0 = 0$, $t_1 = \pi/4$, and $f(t_0, y_0) = 1 \cdot \cos(0) = 1$, we see that

$$y_1 = y_0 + hf(t_0, y_0) = 1 + \frac{\pi}{4}(1) = 1 + \frac{\pi}{4}$$

Step 2. Now that we have y_1 above, we use $t_1 = \pi/4$ and $t_2 = \pi/2$. We then notice that

$$f(t_1, y_1) = \left(1 + \frac{\pi}{4}\right) \cos\left(\frac{\pi}{4}\right) = \left(1 + \frac{\pi}{4}\right) \frac{\sqrt{2}}{2}.$$

Then we compute y_2 in the following manner

$$y_2 = y_1 + hf(t_1, y_1) = \left(1 + \frac{\pi}{4}\right) + \left(1 + \frac{\pi}{4}\right) \frac{\sqrt{2}}{2} \cdot \frac{\pi}{4} = \left(1 + \frac{\pi}{4}\right) \left(1 + \frac{\pi\sqrt{2}}{8}\right) \approx 2.7554.$$

4.c. Directly solving the initial value problem for $y' = y \cos(t)$ follows as

$$\begin{aligned} \frac{dy}{y} &= \cos(t) \\ \int \frac{dy}{y} &= \int \cos(t) dt \\ \ln(y) &= \sin(t) + c \\ y &= e^{\sin(t)+c} = ce^{\sin(t)}. \end{aligned}$$

Then we substitute in $y(0) = 1$ as

$$y(0) = ce^{\sin(0)} = 1 \Leftrightarrow c = 1.$$

Thus, we have that

$$y(t) = e^{\sin(t)}.$$

We were asked to show that $y(t)$ is bounded in the fashion $e^{-\pi} \leq y \leq e^{\pi}$, but $-1 \leq \sin(t) \leq 1$ so naturally $e^{-\pi} \leq e^{-1} \leq y(t) \leq e^1 \leq e^{\pi}$. The rest of (c) we omit for the sake of expedience.

4.d. The backwards Euler formula follows as $y_{n+1} = y_n + hf(t_{n+1}, y_{n+1})$. So we write and solve for y_{n+1} .

$$\begin{aligned} y_{n+1} &= y_n + hy_{n+1} \cos(t_{n+1}) \\ y_{n+1}(1 - h \cos(t_{n+1})) &= y_n \\ y_{n+1} &= \frac{y_n}{1 - h \cos(t_{n+1})} \end{aligned}$$

Furthermore, we are given $h = \pi/4$ with $t_0 = 0$, $y_0 = 1$, $t_1 = \pi/4$, and $t_2 = \pi/2$. So we do our two steps.

Step 1.

$$y_1 = \frac{y_0}{1 - h \cos(t_1)} = \frac{1}{1 - \frac{\pi}{4} \cos\left(\frac{\pi}{4}\right)} = \frac{1}{1 - \frac{\pi\sqrt{2}}{8}}.$$

Step 2.

$$y_2 = \frac{y_1}{1 - h \cos(t_2)} = \frac{\left(\frac{1}{1 - \frac{\pi\sqrt{2}}{8}}\right)}{1 - \frac{\pi}{4} \cos\left(\frac{\pi}{2}\right)} = y_1.$$

So

$$y\left(\frac{\pi}{2}\right) \approx \frac{1}{1 - \frac{\pi\sqrt{2}}{8}}$$

4.e. We ran plots of the Euler's methods and such. The code can be found in:

<https://github.com/chtompki/MathSpring2026/tree/main/NumericalAnalysis/ProblemSet4>

The exact solution for the problem can be found back in part (c) of the problem. That said, we have generated a plot of the forward and backwards Euler's method given in Figure

1

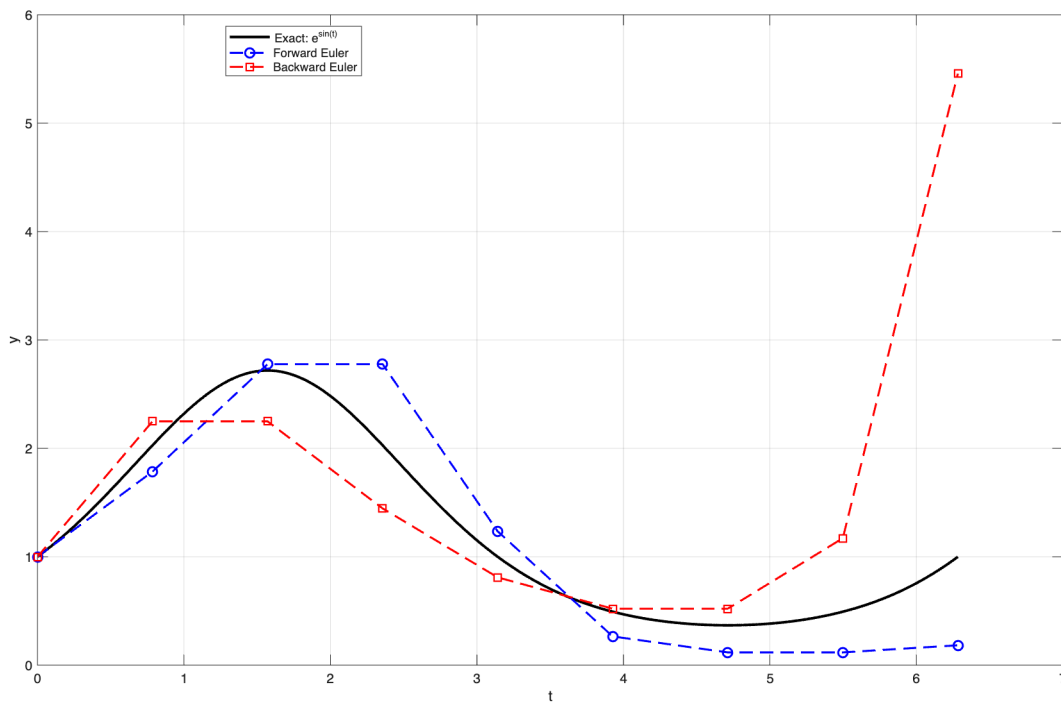


FIGURE 1. For $y' = y \cos(t)$, a plot of the exact solution along with Euler's backwards and forwards methods.

We've omitted talking about the Global Error Theorem for Euler's method out of the need for expedience.

5. PROBLEM 5